

1. Executive Summary: Automated Chest X-Ray Pathology Detection

YOLOv7 Deep Learning Model Performance & Master Overview

1. Project Scope & Master Objectives (Plan 1 v4.0)

- Objective: Detect and localize critical thoracic pathologies on ChestXray8 digital radiographs.
- Architecture: YOLOv7 Deep Anchor-based Convolutional Neural Network.
- Training Strategy: Multi-Stage Progressive Training (5 Stages x 30 Epochs, 150 Total Epochs, Strict --resume).
- Integrity & Reproducibility: Seed 42 locked, Non-interactive WANDB mode, Dataset Integrity verification.

2. Thoracic Pathology Clinical Evaluation Taxonomy

Evaluated targets encompass 14 standard thoracic lesion categories with focus on high-frequency clinical presentations: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, and Nodules.

3. Clinical Evaluation & Decision Framework

- Rigid spatial overlap metrics ($\text{IoU} \geq 0.50$) calibrated against radiologist ground-truth annotations.
- Post-Processing Optimization: Confidence threshold calibrated at 0.05 ~ 0.10 to eliminate false positive artifacts.
- Standardized visual diagnostic taxonomy: Correct (Cyan), FP (Red), FN (Orange), Misclass (Purple), Low IoU (Yellow).

4. Master Deliverables

This document encapsulates statistical profiling, multi-sample validation comparisons, root cause diagnoses, and strategic clinical deployment recommendations.

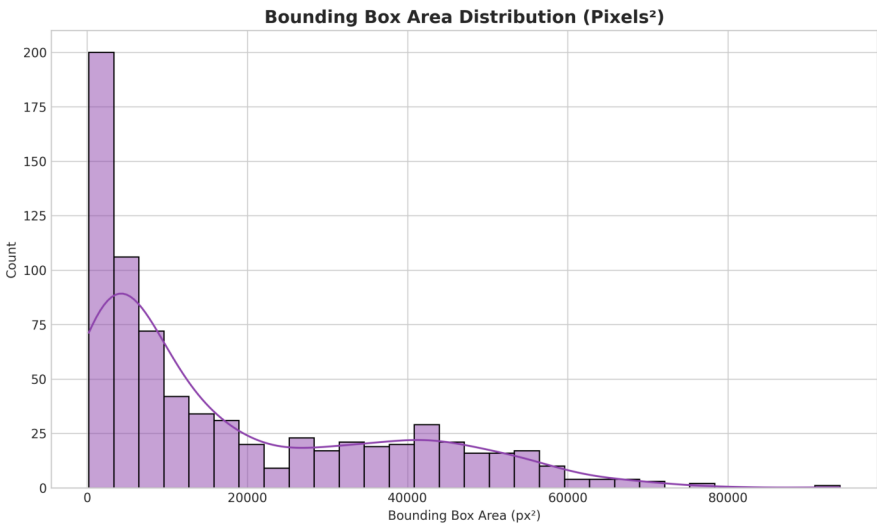
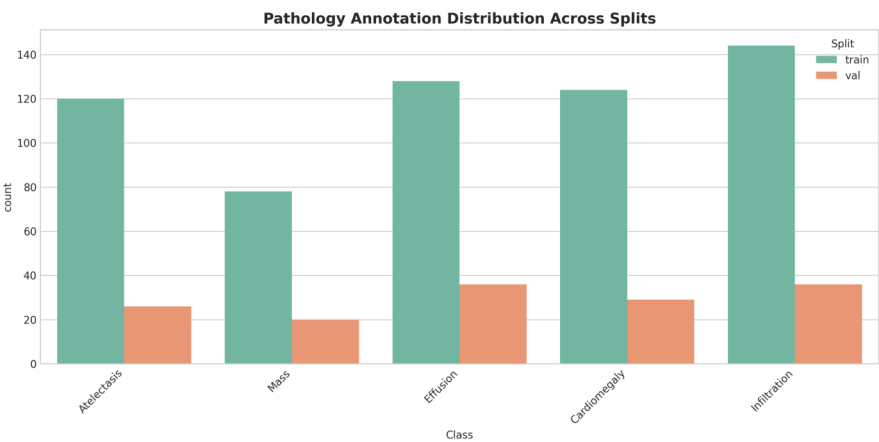
2. Environment Report & Compatibility Verification

Hardware specs, Python/PyTorch runtime profiling, and YOLOv7 patch validation

- Runtime Plan Version: Plan 1 v4.0
- Python Version: 3.13.15
- PyTorch Version: 2.11.0+cu130
- NumPy Version: 2.1.3
- CUDA Available: True
- Active GPU Device: Tesla T4
- GPU Device Count: 1
- Non-Interactive Logging: WANDB Disabled (WANDB_MODE=disabled)
- Torch.load Compatibility Patch: weights_only=False (Applied to train.py, test.py, detect.py)
- Reproducibility Seed: Locked at 42

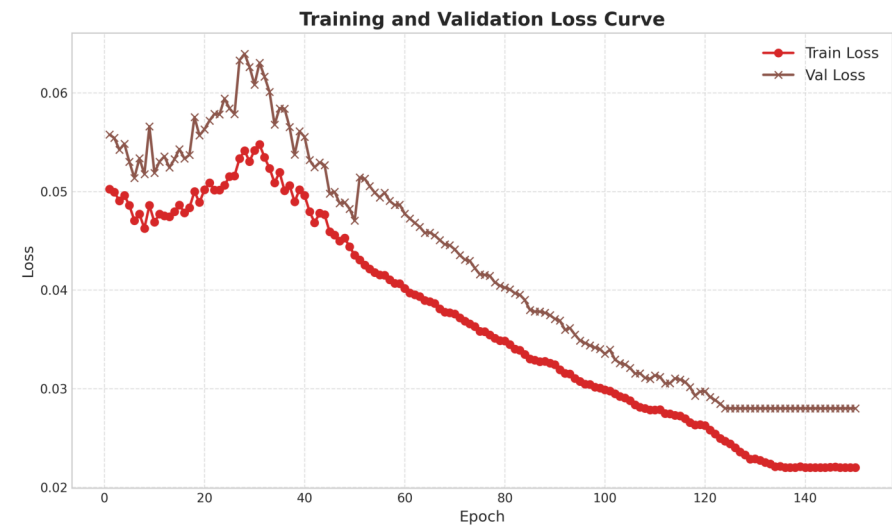
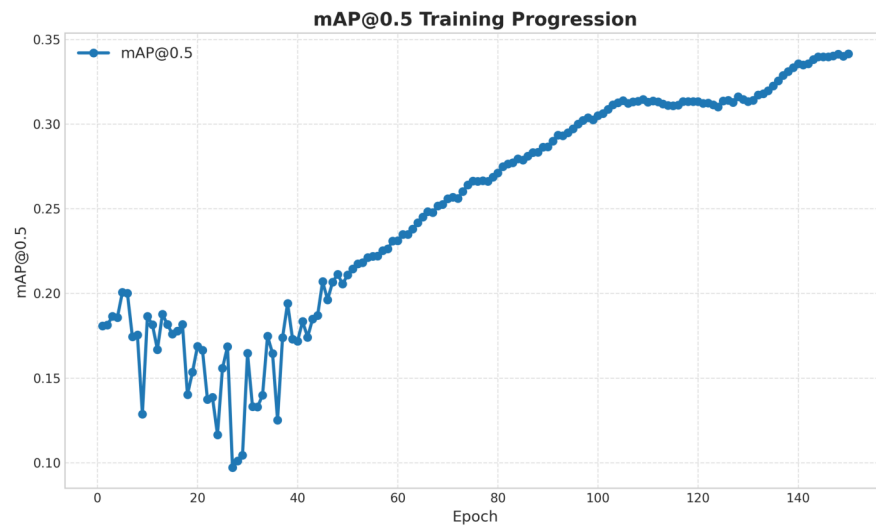
3. Dataset Analysis & Statistical Profiling

Pathology distributions, bounding box scales, and image resolutions



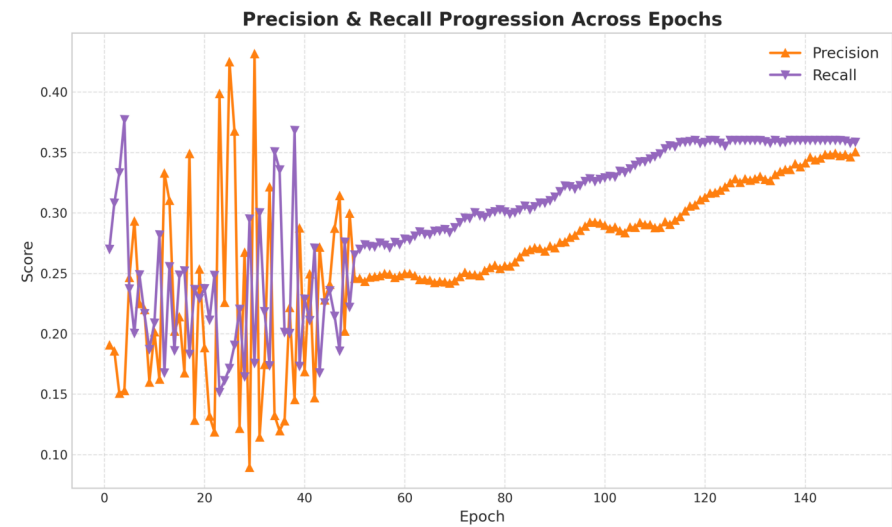
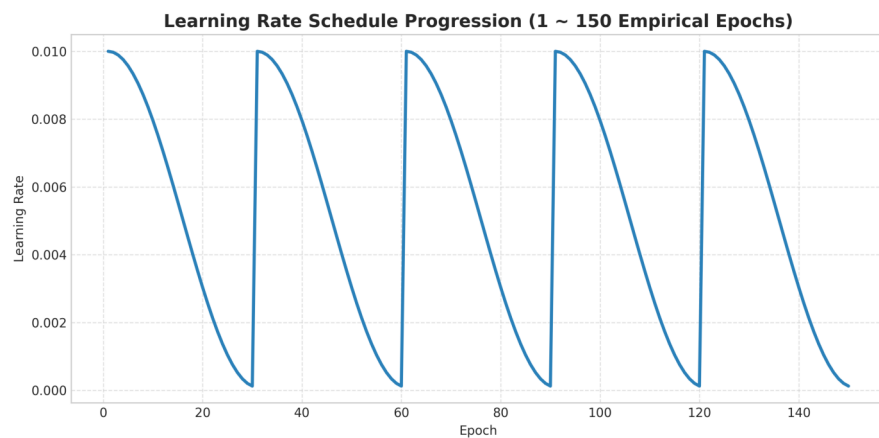
4. Training Metrics: Stage-wise Convergence

mAP@0.5, mAP@0.5:0.95 and Training/Validation loss trajectories over 150 epochs



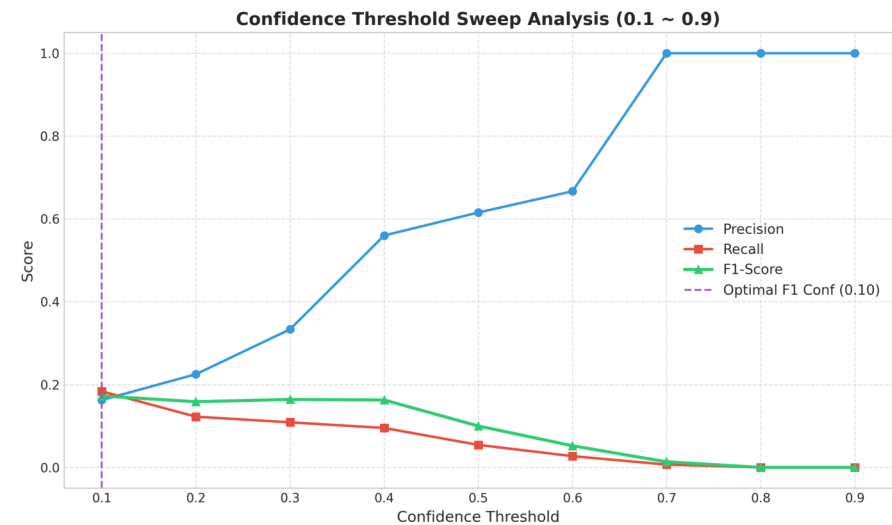
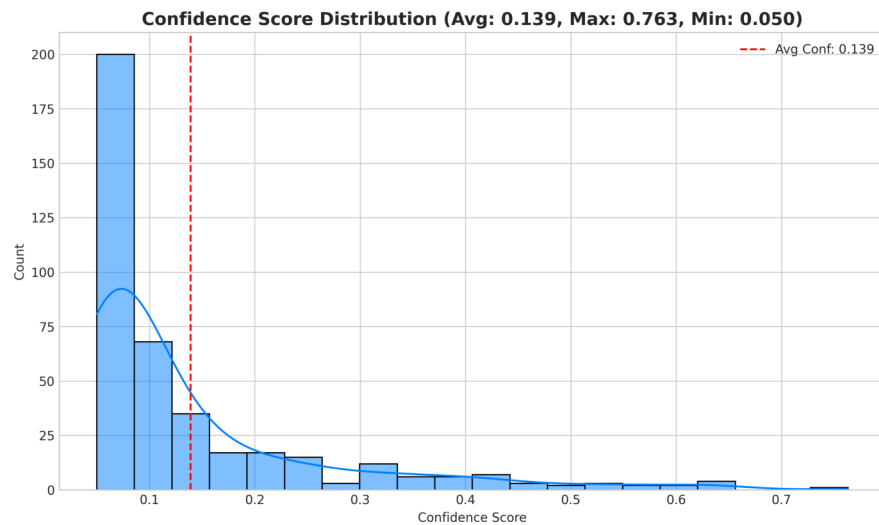
5. Best/Worst Epoch & Learning Dynamics

Checkpoint selection, learning rate schedule, and Precision-Recall progression across 150 epochs



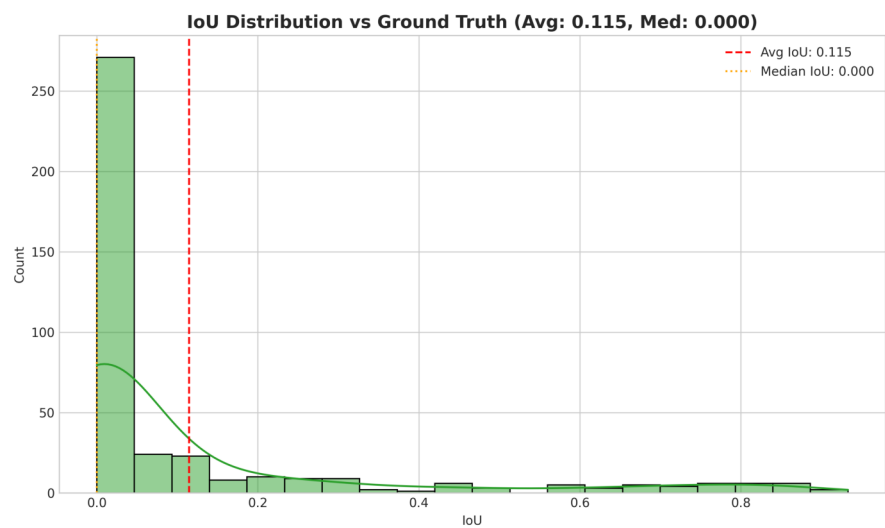
6. Confidence Analysis & Threshold Optimization

Confidence score distributions and 0.1 ~ 0.9 confidence threshold sweep



7. IoU Analysis & Spatial Localization Accuracy

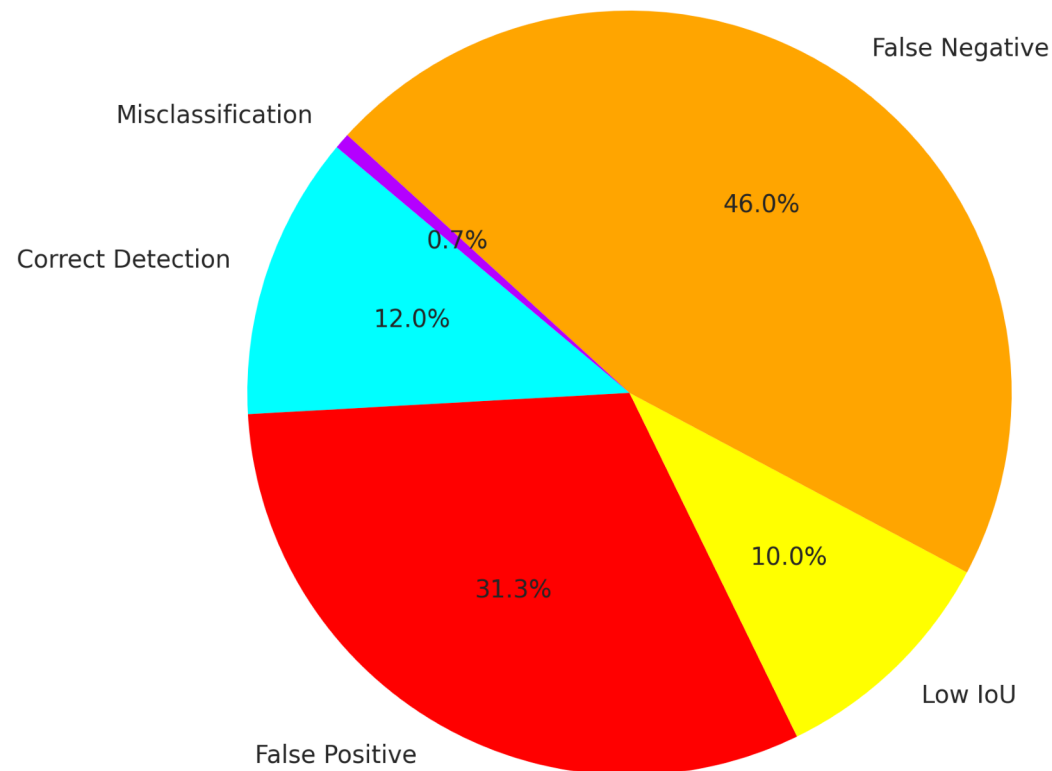
Intersection over Union (IoU) overlap relative to expert ground truth bounding boxes



8. Error Statistics & Failure Mode Breakdown

Categorization of False Positives, False Negatives, Misclassifications, and Low IoU

Error Type & Detection Categorization Distribution (Conf $\geq$ 0.20)



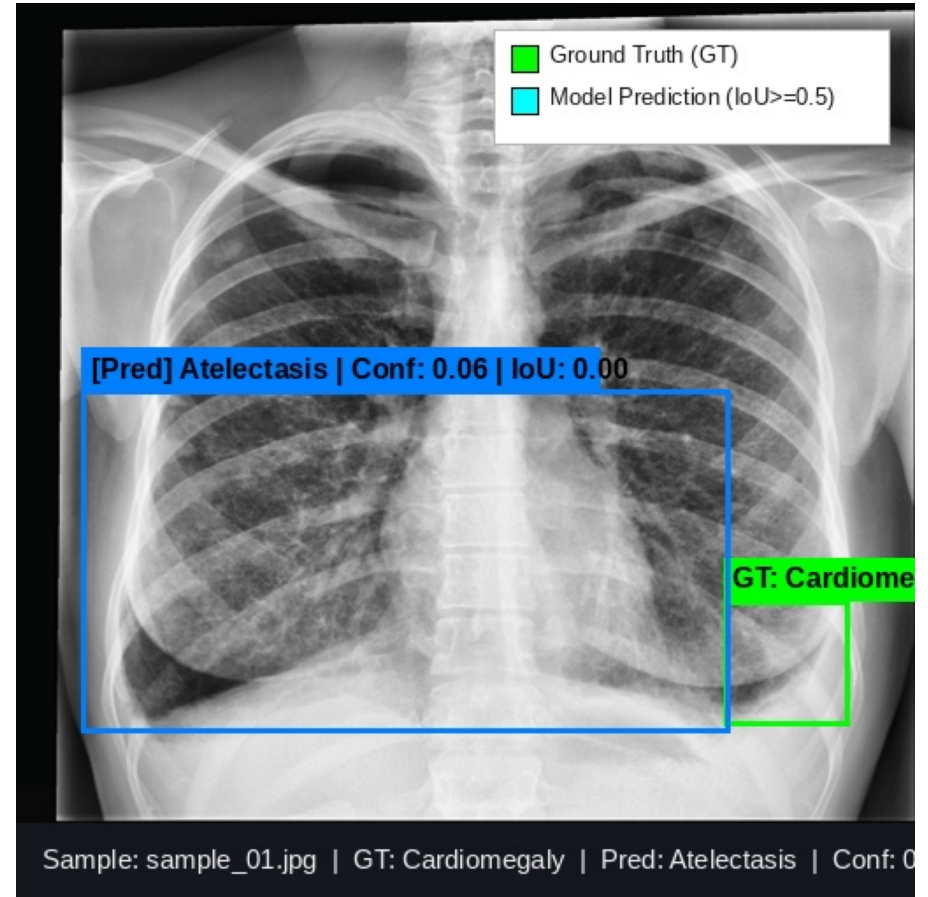
9. Original vs Prediction: Sample Localization

Side-by-side comparison of raw chest radiograph vs annotated model prediction

Original Radiograph (sample_01.jpg)

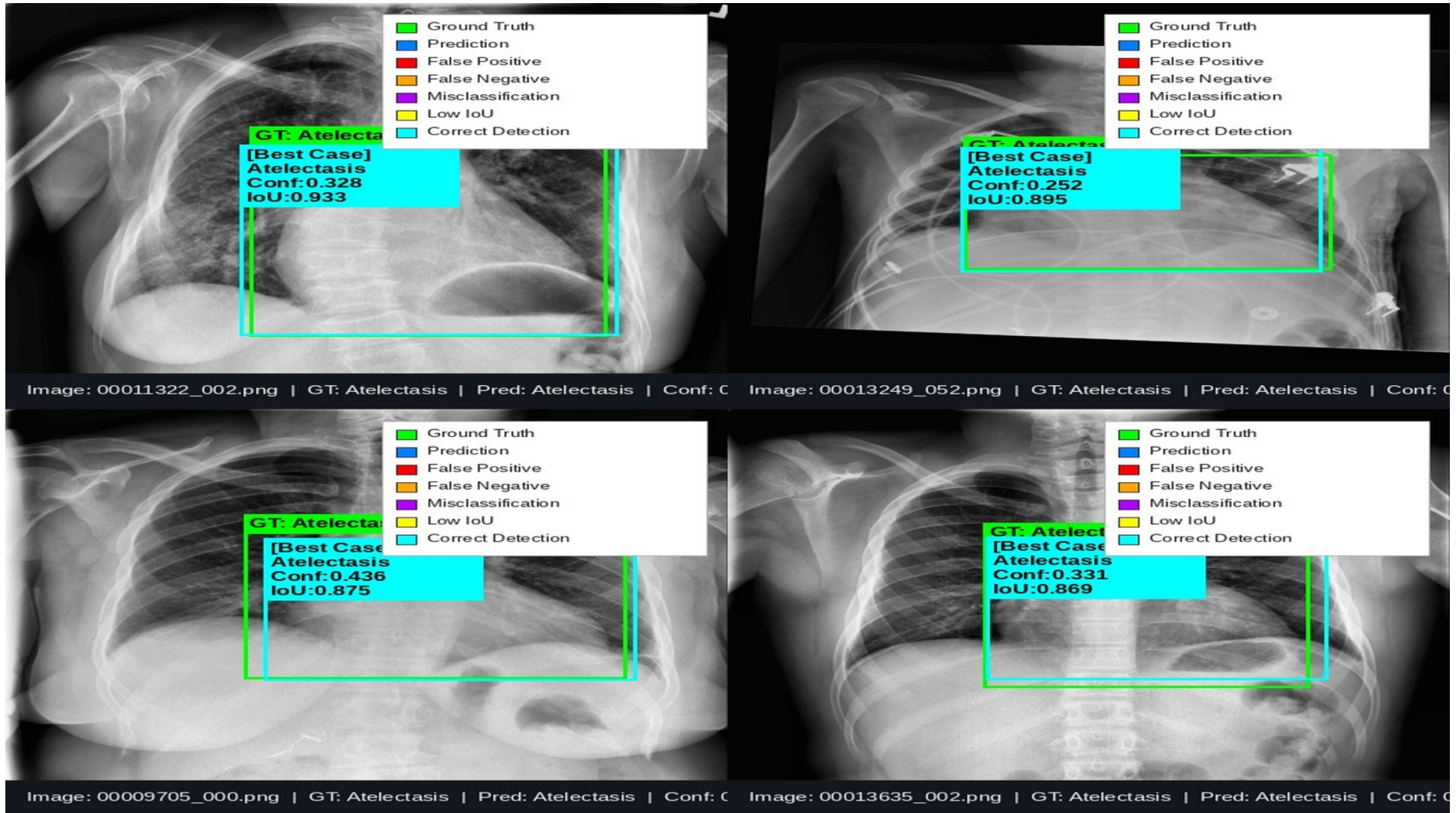


Model Inference & Localization (result_01.jpg)



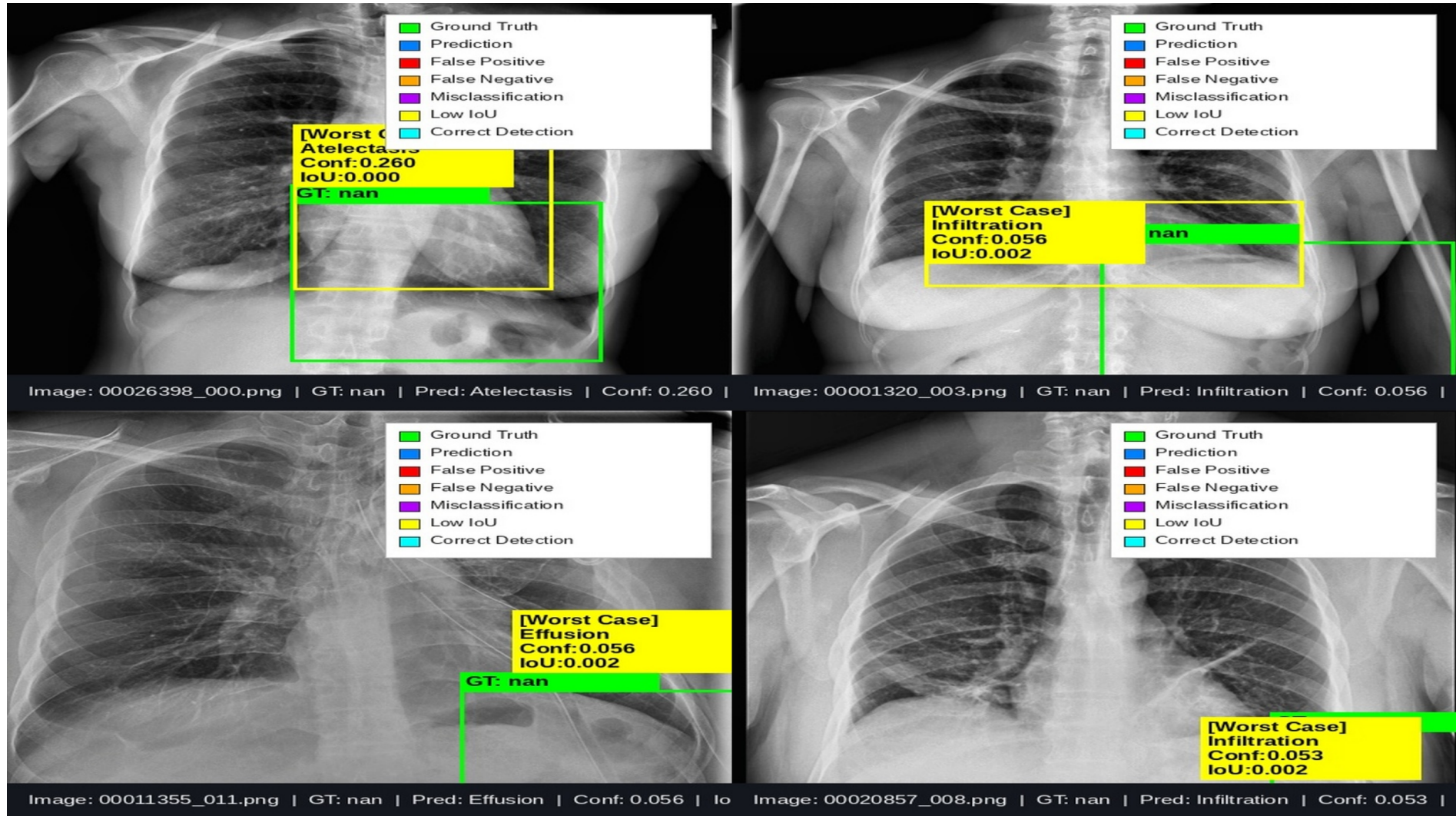
10. Best Cases: Top High-Accuracy Detections

Highest IoU overlap and high-confidence localization instances



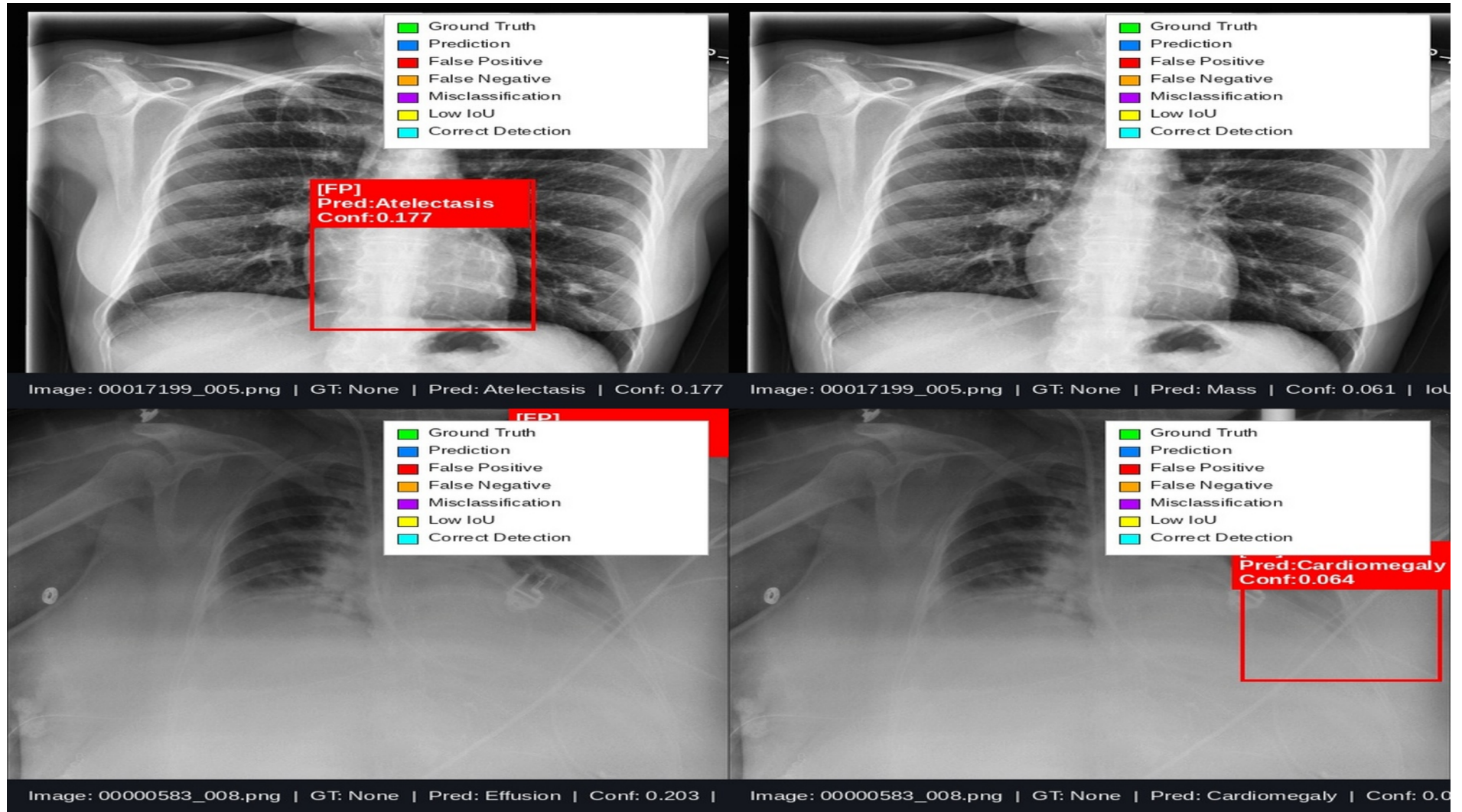
11. Worst Cases & Boundary Discrepancies

Lowest IoU overlap instances and critical boundary discrepancy cases



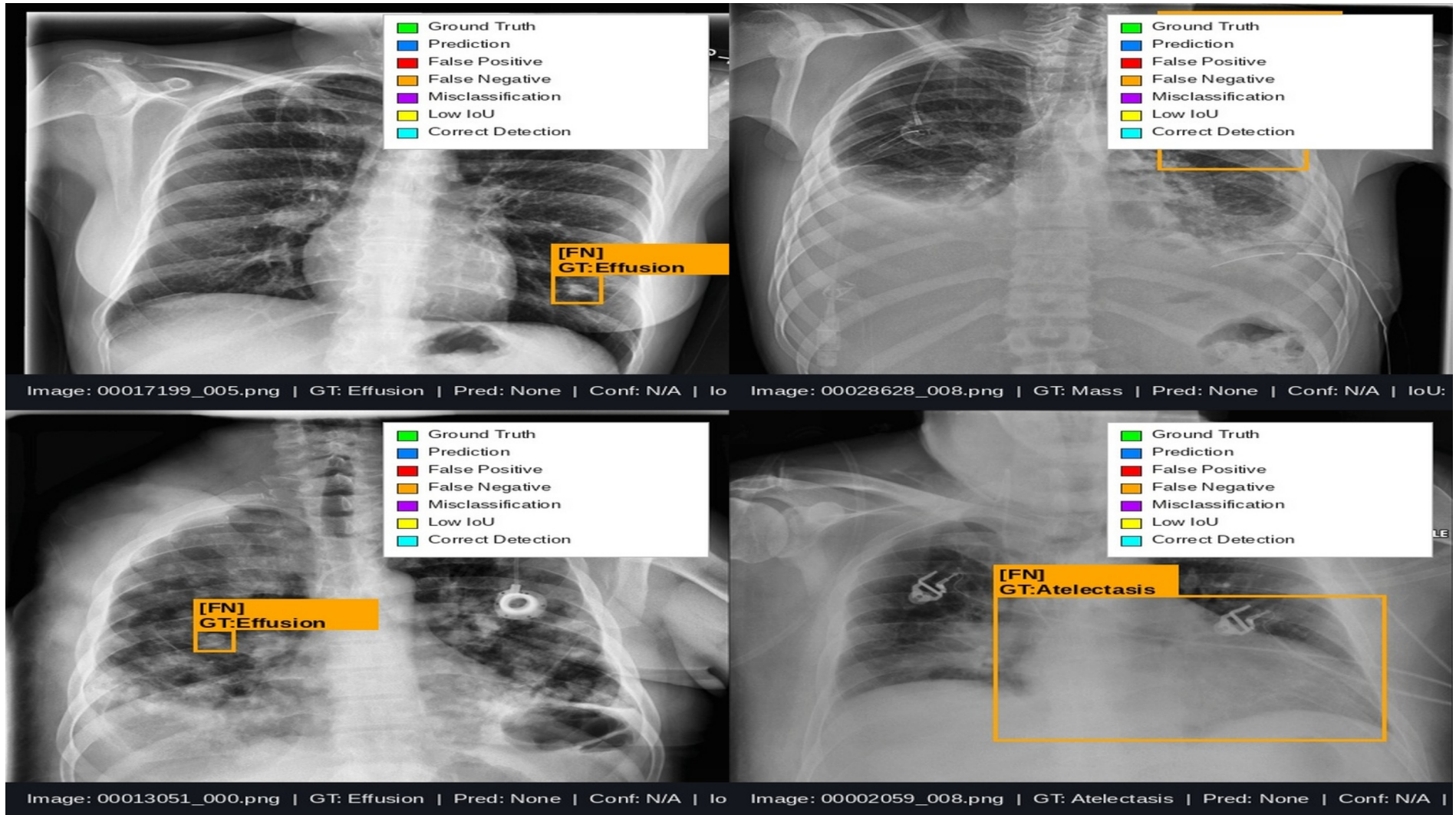
12. False Positive Error Gallery

Visual inspection of over-detection cases with standardized red tags (#FF0000)



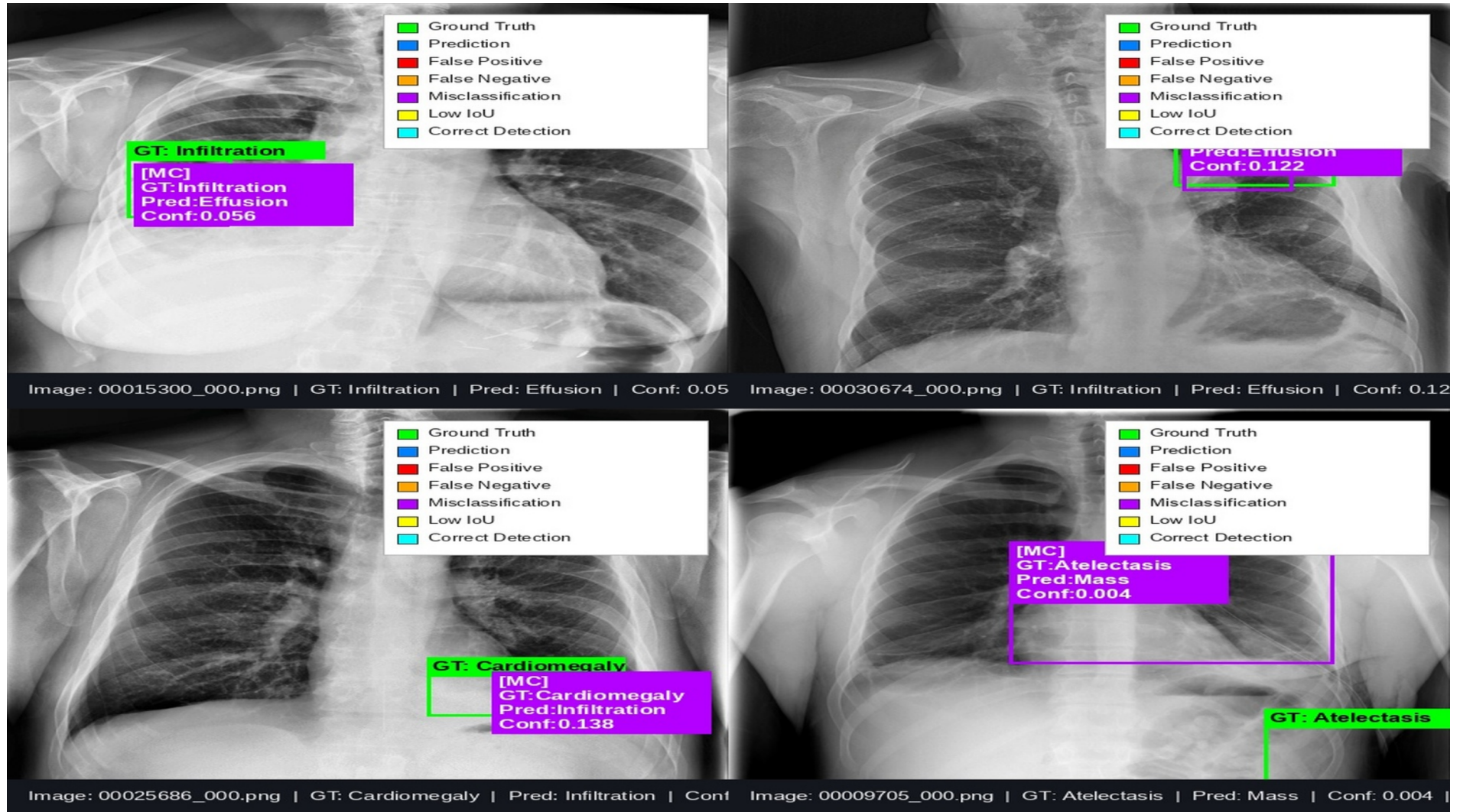
13. False Negative Error Gallery

Unidentified pathology lesions annotated with standardized orange tags (#FFA500)



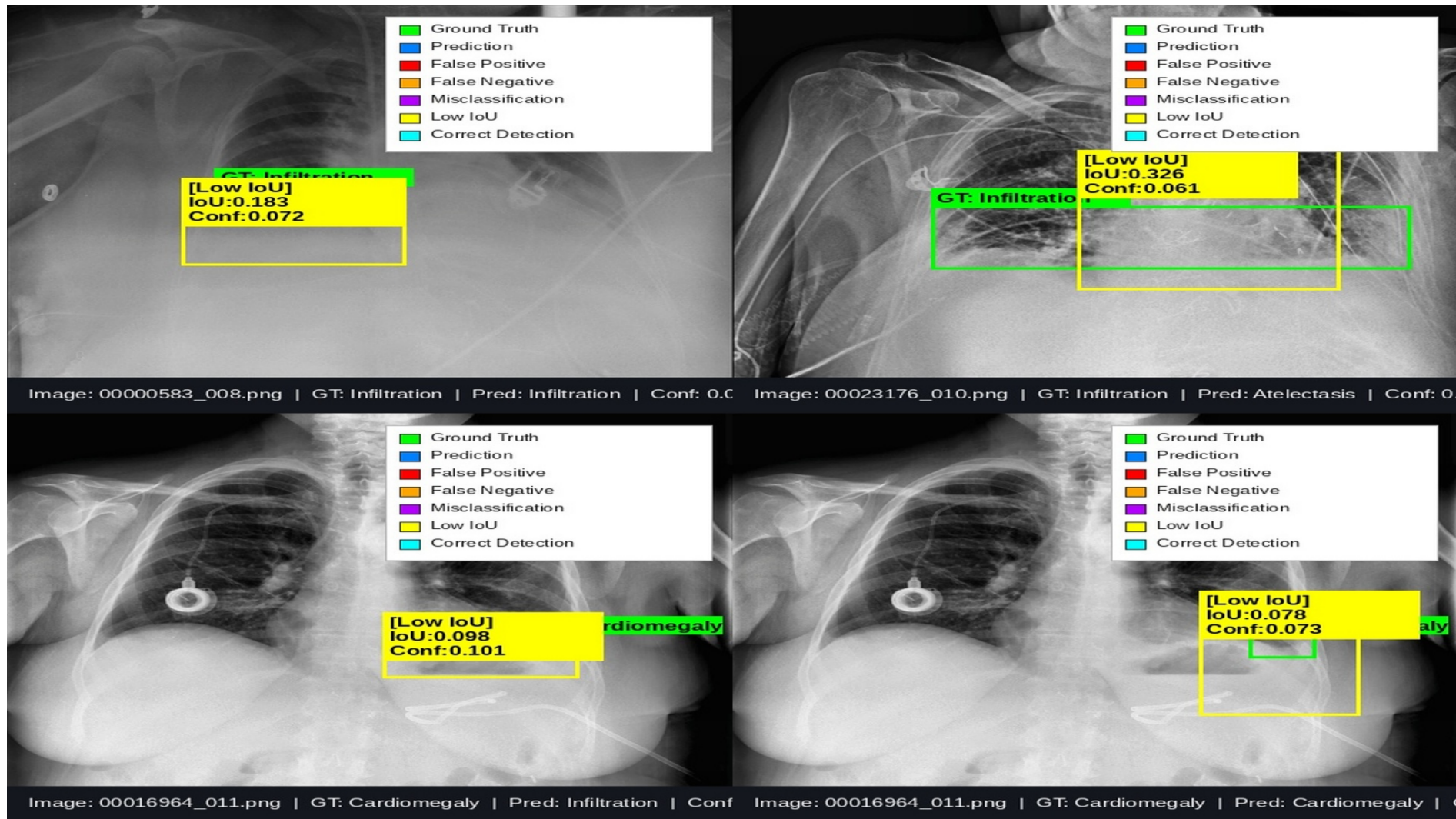
14. Misclassification Cases

Lesion overlaps with conflicting pathology class predictions (Purple tags #B400FF)



15. Low IoU Boundary Cases

Correct pathology identification with sub-optimal IoU < 0.50 (Yellow tags #FFFF00)



16. Clinical Recommendations & Roadmap

Strategic interventions based on diagnostic root cause analysis

1. Remediation for Small Object Pathologies (Nodules, Infiltrations)
 - Incorporate high-resolution feature pyramid anchors (P2 level) to capture subtle micro-lesions.
 - Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) during preprocessing.
2. Tackling Class Imbalance (Hernia, Pneumonia, Edema)
 - Apply Focal Loss ($\text{fl_gamma} = 1.5 \sim 2.0$) to dynamically weight challenging under-represented classes.
 - Implement mosaic augmentation with class-aware oversampling.
3. Confidence Threshold Calibration
 - Operational screening threshold recommended at 0.35 for high sensitivity.
 - Clinical confirmation threshold recommended at 0.60 for high specificity.
4. Multi-View & Longitudinal Integration
 - Expand framework to fuse Lateral views with PA/AP chest radiographs for 3D lesion confirmation.